

A**DATA PREPARATION***(outside CV, label-free)***Raw aCGH Data**

100 samples x 2,834 regions
 Discrete CN: {-1, 0, +1, +2}

**Region Merging**

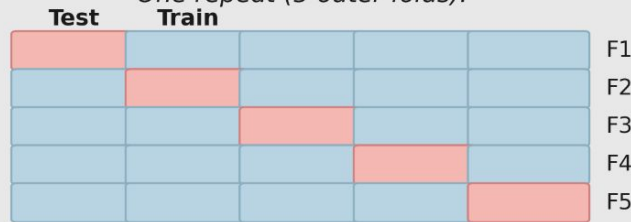
Pearson $r > 0.8$ threshold
 Label-free (no target leakage)

**Merged Data**

100 samples x 273 segments
Input to nested CV

**B****REPEATED STRATIFIED 5-FOLD CV**

200 repeats (seeds 1001 - 1200)

One repeat (5 outer folds):**Within each outer fold:****80% Train**

Fit hierarchical
 classifier

20% Test

Evaluate:
 BA (3-class)

5-Fold Inner CV

Tunes Stage 2
Scoring: balanced acc.

Primary metric:
 Stage 2 BA (BA2)

200 repeats x 5 folds = 1,000 estimates*Report: mean +/- std***C****HIERARCHICAL CLASSIFIER****Test samples****STAGE 1**

HER2+ vs Rest (binary)

BA=1.0

KW top-5 + Random Forest
 500 trees, balanced weights
FIXED across all pipelines

HER2+**HER2+****Rest****STAGE 2**

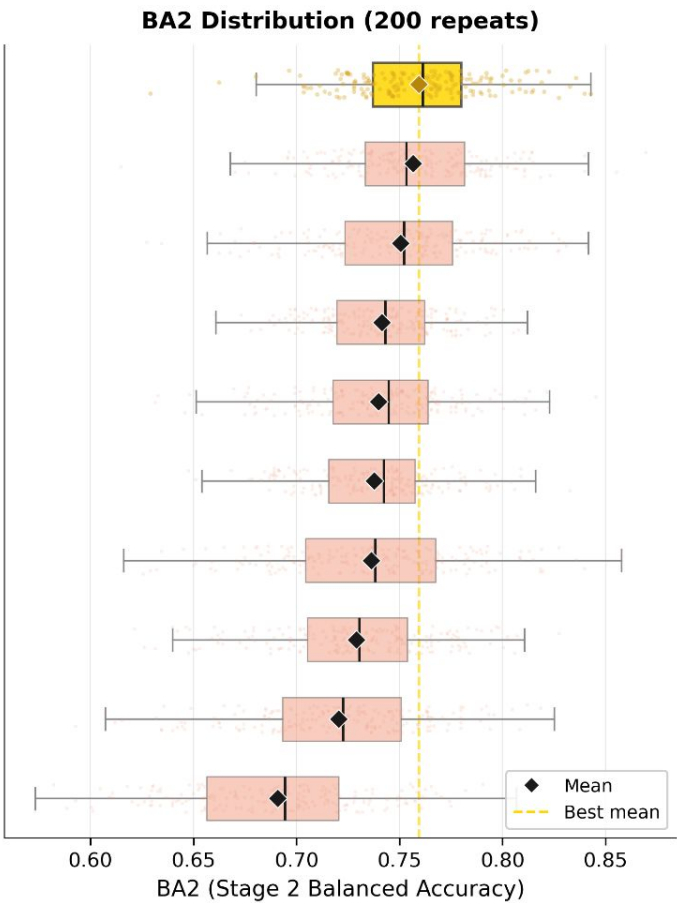
HR+ vs Triple Neg (binary)

Feature selection + Classifier
TUNED via inner CV
 4 pipeline variants compared

HR+**Triple Neg****3-class prediction**

#	Pipeline	BA2 (mean +/- std)	Combined BA (mean +/- std)	AUROC	Acc.	Feat.
1	EN + NMC (plat.) *	0.759 +/- 0.032	0.840 +/- 0.022	0.951	0.837	82.3
2	NMC Pens Ens.	0.757 +/- 0.037	0.838 +/- 0.025	0.950	0.835	-
3	KW + NMC (plat.)	0.750 +/- 0.040	0.834 +/- 0.026	0.946	0.831	100.0
4	Standalone EN (plat.)	0.741 +/- 0.029	0.828 +/- 0.019	0.944	0.826	-
5	NMC Ensemble	0.740 +/- 0.037	0.827 +/- 0.025	0.941	0.823	-
6	Standalone EN	0.738 +/- 0.033	0.825 +/- 0.022	0.938	0.823	-
7	KW + NMC	0.736 +/- 0.046	0.824 +/- 0.031	0.938	0.821	41.6
8	EN + NMC	0.729 +/- 0.040	0.819 +/- 0.026	0.935	0.816	32.5
9	KW + RF	0.721 +/- 0.044	0.814 +/- 0.029	0.932	0.810	46.9
10	EN + RF	0.691 +/- 0.046	0.794 +/- 0.031	0.919	0.791	32.4

* Submitted method



BA2 = Stage 2 balanced accuracy (HR+ vs TN). Combined BA includes perfect Stage 1 (HER2+ vs rest). 200 repeated stratified 5-fold CV.

Raw aCGH Data

100 samples
2,834 regions
CN: {-1,0,1,2}

$n=100$
 $p=2,834$

Region Merging

Label-free
Pearson $r > 0.8$
Adjacent collapse

2,834 $\rightarrow$ 273
 $n=100$
 $p=273$

Flat: KW+RF

True	Predicted		
	HER2+	HR+	TN
HER2+	0.99	0.00	0.01
HR+	0.01	0.74	0.26
TN	0.00	0.35	0.64



Hierarchical: EN+NMC(P)

True	Predicted		
	HER2+	HR+	TN
HER2+	1.00	0.00	0.00
HR+	0.00	0.77	0.23
TN	0.00	0.26	0.74

Recall (row-normalized)

**Stage 1:
HER2+ vs Rest**

KW top-5 + RF (500 trees)
Balanced weights
FIXED (no tuning)

BA = 1.0

5 chr17 features (ERBB2)
 $k=5$
features



Rest

**Stage 2:
HR+ vs TN**

Elastic Net selector
 $C=0.046$, $l1=0.1$, $k=50$
NMC (no shrinkage)
Plateau ensemble:
top 15 configs pooled

~82 features/fold
chr6p, chr12q, chr5q, chr16, chr22
~82 features

HER2+

Predicted
HER2+

HR+

TN

3-Class Prediction

HER2+
HR+
Triple Neg

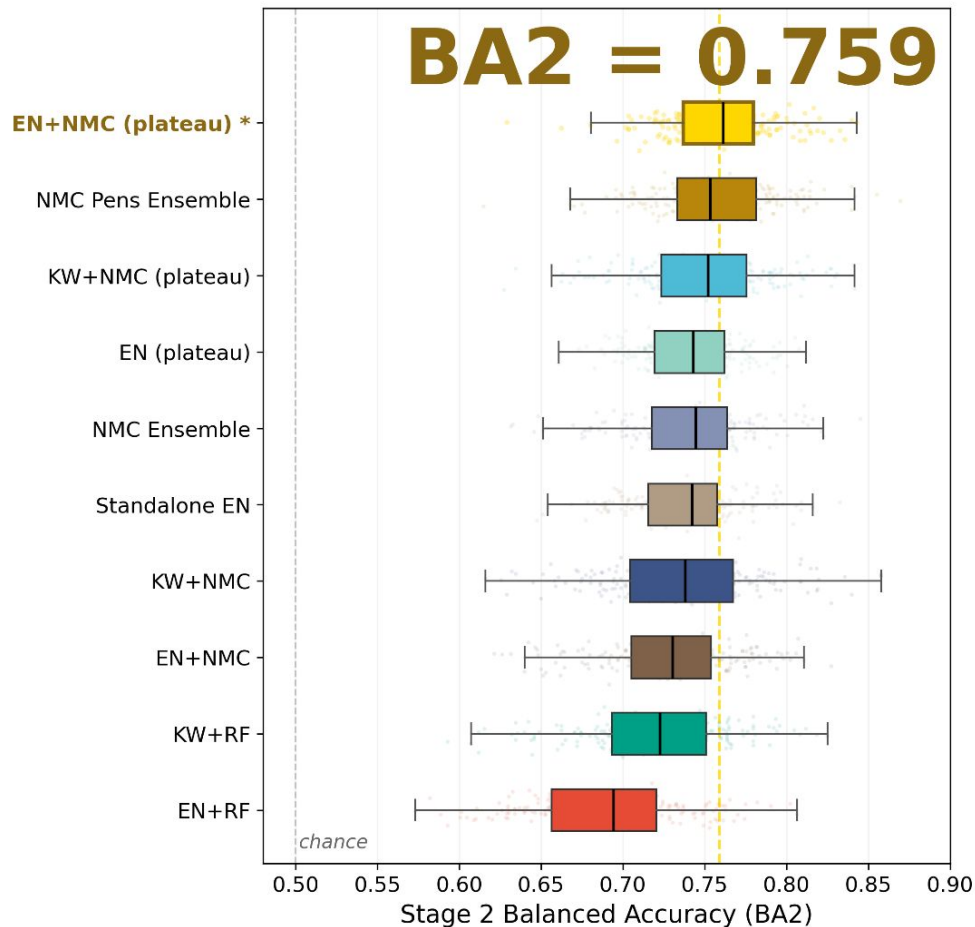
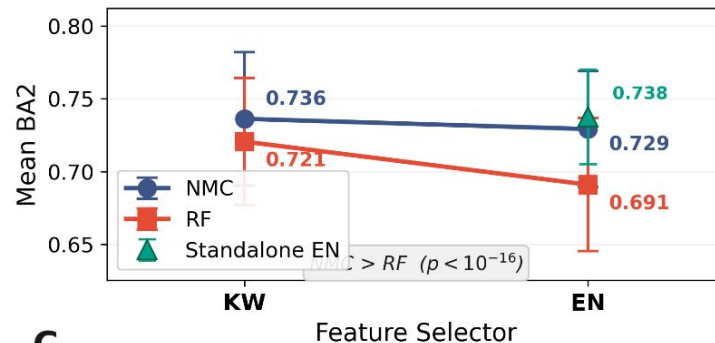
Estimate:

48/57 correct (~84.2%)

Preprocessing

Stage 1 (HER2+ vs Rest)

Stage 2 (HR+ vs TN)

A**B****C**

Why EN+NMC (plateau)?

1. Simple classifier wins.

NMC > RF across all selectors ($p < 10^{-16}$).
Fewer parameters resist overfitting ($n = 72$).

2. Plateau exposes true EN+NMC quality.

Base EN+NMC (0.729) < standalone EN (0.738): inner CV noisy at small n . Plateau pools top configs $\rightarrow$ 0.759.

3. EN+NMC(P) outperforms standalone EN(P).

Best model overall (0.759 vs 0.741): separating selection from classification yields higher BA2.

4. Aligns with Wessels et al. (2005).

Simple classifiers with univariate filtering match or exceed complex pipelines on small- n discrete data.